Teaching Neural Inertial Odometry with Supervisory Measurements

Anonymous Author(s)
Affiliation
Address
email

Abstract: Neural inertial odometry has shown strong potential for motion estimation in challenging environments, but most existing methods rely on dense ground-truth trajectories for supervision, which are often unavailable in real robotic deployments. We propose Supervisory Measurement Guided Neural Inertial Odometry, a framework that learns inertial odometry models using sparse but reliable supervisory measurements instead of full trajectory supervision. The key idea is to leverage high-confidence observations, such as loop closures or precise relative pose sources, to guide the learning of neural motion representations. The proposed method integrates neural inertial prediction with a probabilistic filtering framework and formulates training as a bilevel optimization problem that separates trajectory inference from parameter learning. Experiments on simulated and real-world datasets demonstrate that the proposed approach improves odometry accuracy and robustness compared to existing neural inertial odometry baselines.

Keywords: Neural inertial odometry, probabilistic factorization, bilevel optimization

1 Introduction

2 Related Works

Table 1: Comparison of related methods.

Algorithm	Network	Loss	Input	Output

3 Preliminaries

3.1 Matrix Lie Group

Let $\mathbf{G}$ be a matrix Lie group with Lie algebra $\mathfrak{g}$. Note that $\mathfrak{g}$ is isomorphic to $\mathbb{R}^{\dim \mathfrak{g}}$. The exponential map $\exp : \mathbb{R}^{\dim \mathfrak{g}} \to \mathbf{G},\ \xi \mapsto \exp_{\mathrm{m}}(\xi^{\wedge})$ relates $\mathfrak{g}$ to its associated group $\mathbf{G}$, where $(\cdot)^{\wedge}$ denotes the skew operator and $\exp(\cdot)$ the matrix exponential. In a neighborhood of the identity, $\exp$ is locally invertible, enabling the definition of the logarithm map $\log : \mathbf{G} \to \mathbb{R}^{\dim \mathfrak{g}},\ X \mapsto \log(X)^{\vee}$, where $(\cdot)^{\vee}$ is the inverse of $(\cdot)^{\wedge}$. The special orthogonal group $SO(3)$ represents the set of all possible rotations of a rigid body in three-dimensional space. The special Euclidean group $SE(3)$ (rigid-body transformations), and $SE_2(3)$ are defined as

$$SE(3) \triangleq \left\{ \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \,\middle|\, R \in SO(3), p \in \mathbb{R}^3 \right\}, SE_2(3) \triangleq \left\{ \begin{bmatrix} R & v & p \\ \mathbf{0} & & I \end{bmatrix} \middle| R \in SO(3), p, v \in \mathbb{R}^3 \right\},$$

where $\mathbf{0}$ and I denote the zero matrix and the identity matrix, with the dimension omitted when clear from context. For $x^{\wedge}, y^{\wedge} \in \mathfrak{g}$ and $\|x\|$ small, their compounded exponentials can be approximated

Submitted to the 10th Conference on Robot Learning (CoRL 2026). Do not distribute.

to $\exp(x)\exp(y) \approx \exp(\mathrm{dexp}_y^{-1} x + y)$, where dexp_x is the left Jacobian of x. The adjoint operator has the property that $\mathrm{Ad}_{X^{-1}} = (\mathrm{Ad}_X)^{-1}$.

Let $\mathcal{M}$ be the smooth manifold of a Lie group. For any $X_1, X_2 \in \mathcal{M}$ and tangent vector $\xi \in \mathbb{R}^{\dim \mathcal{M}}$, we define the *plus* and *minus* operators, $X_1 \boxplus \xi$ and $X_1 \boxminus X_2$, which map between the manifold and its tangent space. These operators admit left and right versions; the choice is application-dependent and left implicit. For instance, in the left case, $X_1 \boxplus \xi = \exp(\xi) X_1, X_1 \boxminus X_2 = \log(X_1 X_2^{-1})$. On product manifolds, the operators act componentwise. The uncertainty of $X \in \mathcal{M}$ is modeled in the tangent space by a Gaussian distribution. Let $\xi \sim \mathcal{N}(0, \Sigma)$, the induced distribution is

$$\mathcal{N}_L(\bar{X}, \Sigma) \triangleq \eta \exp\left(-\tfrac{1}{2}(X \boxminus \bar{X})^\top \Sigma^{-1} (X \boxminus \bar{X})\right), \tag{1}$$

where η is the normalization coefficient.

3.2 IMU Kinematics

The kinematics of the state is given by

$$\dot{R}_t = R_t\, \omega_t^\wedge, \quad \dot{p}_t = v_t, \quad \dot{v}_t = a_t. \tag{2}$$

The increments of the poses are given by IMU, whose measurement model is

$$\tilde{\omega}_t = \omega_t + b_t^g + n_g, \quad \tilde{a}_t = R_t^\top (a_t - g) + b_t^a + n_a, \tag{3}$$

where b_t^g and b_t^a denote the gyroscope and accelerometer bias terms, respectively, n_g and n_a are the corresponding white Gaussian noises, and $\tilde{\cdot}$ denotes the sensor measurements. It is noted that when the state is defined on a Lie group, the above dynamics satisfy a group-affine property.

Determining appropriate values for b_t^a and b_t^g remains a long-standing challenge. Typically, the biases are modeled as a Brownian motion as

$$\dot{b}_t^g = n_{b_g}, \quad \dot{b}_t^a = n_{b_a}, \tag{4}$$

where n_{b_g} and n_{b_a} denote the white noises.

3.3 Invariant Observer

Consider the group-affine systems

$$\dot{X}_t = f_{u_t}(X_t), \tag{5}$$

where u_t is an input variable. Consider two distinct trajectories X_t and $\bar{X}_t$ of (5). The left-invariant and right-invariant errors between the two trajectories are defined by

$$\eta_t = X_t^{-1} \bar{X}_t \ \text{ or } \ \bar{X}_t^{-1} X_t \ \text{ (left invariant)},$$
$$\eta_t = X_t \bar{X}_t^{-1} \ \text{ or } \ \bar{X}_t X_t^{-1} \ \text{ (right invariant)}.$$

According to Theorem 1 in [1], a well-known and useful finding is that the invariant error is state trajectory independent. Let $\xi_t \triangleq \log(\eta_t)$, and it follows the linear differential equation in $\mathbb{R}^d$:

$$\dot{\xi}_t = A_t \xi_t + B_t n_t, \tag{6}$$

where $n_t = [n_t^a; n_t^g]$.

3.4 Continuous Adjoint Method for NODEs

Consider a Neural ODE

$$\dot{x}(t) = f\big(x(t), u(t); \theta\big), \qquad t \in [t_0, t_N], \tag{7}$$

with initial condition $x(t_0)$. Assume the loss depends on the discrete-time states along the trajectory, i.e., $L = L\big(x(t_0), \ldots, x(t_N)\big)$. To compute $\frac{dL}{d\theta}$ efficiently, the continuous adjoint method introduces the adjoint variable $\alpha(t) \triangleq \frac{\partial L}{\partial x(t)}$ and solves it backward in time. The dynamics of $\alpha(t)$ is

$$\dot{\alpha}(t) = -\left(\frac{\partial f}{\partial x}\big(x(t), u(t); \theta\big)\right)^\top \alpha(t). \tag{8}$$

If the loss contains discrete terms at times $\{t_k\}$, the adjoint satisfies the jump condition $\alpha(t_k^-) = \alpha(t_k^+) + \frac{\partial L}{\partial x(t_k)}$. The gradient with respect to θ is then given by

$$\frac{dL}{d\theta} = \int_{t_0}^{t_N} \alpha(t)^\top \frac{\partial f}{\partial \theta}\big(x(t), u(t); \theta\big)\, dt. \tag{9}$$

4 Problem Formulation

Following Liu et al. [2], we consider b_t^g and b_t^a to be **deterministic** variables with dynamics modeled by a NODE, as follows:

$$\dot{b}_t^g = f_g(b_t^g, \tilde{\omega}_t; \theta), \quad \dot{b}_t^a = f_a(b_t^a, \tilde{a}_t; \theta), \tag{10}$$

where θ denotes the network parameters to be trained.

The tasks of this paper are summarized as follows:

1. Learn the dynamics of b_t^a and b_t^g in (10).
2. Learn the covariances of n_a, n_g.

5 Supervisory Learning Objectives

Given the bias dynamics in (10), the network parameters θ and the IMU noise statistics cannot be learned from direct bias supervision, since the bias trajectory is typically unavailable in practice. Following [2], we instead learn them indirectly from supervisory measurements defined on the propagated motion states.

5.1 Marginal-Likelihood Formulation

Let Y denote the stacked supervisory measurements. We formulate parameter learning as the maximization of the marginal likelihood

$$\hat{\theta} = \arg\max_{\theta \in \Theta} p(Y \mid \theta). \tag{11}$$

To evaluate this likelihood, we introduce the latent trajectory states $X_{1:N}$ generated by lower-level IMU propagation under the learned bias dynamics. Their evolution is modeled as

$$X_{k+1} = f_k(X_k; \theta) + w_k, \qquad w_k \sim \mathcal{N}(\mathbf{0}, Q_k), \tag{12}$$

where Q_k denotes the propagation covariance induced by IMU noise.

Since supervisory measurements are only available at selected time indices, we partition the latent trajectory into two subsets: the supervisory states $\boldsymbol{X}^s$ and the remaining odometry states $\boldsymbol{X}^o$. Then

$$p(Y \mid \theta) = \int p(Y \mid \boldsymbol{X}^s)\, p(\boldsymbol{X}^s \mid \theta)\, \mathrm{d}\boldsymbol{X}^s, \tag{13}$$

where

$$p(\boldsymbol{X}^s \mid \theta) = \int p(X_{1:N} \mid \theta)\, \mathrm{d}\boldsymbol{X}^o. \tag{14}$$

The exact marginalization over $\boldsymbol{X}^o$ is generally intractable. We therefore approximate $p(\boldsymbol{X}^s \mid \theta)$ by a Gaussian distribution obtained from lower-level IMU propagation, with mean $\bar{\boldsymbol{X}}^s$ and covariance $\bar{\boldsymbol{P}}^s$.

5.2 Mean Pretraining and Covariance-Aware Refinement

Let the supervisory residual on the manifold be

$$r = Y \boxminus h(\bar{\boldsymbol{X}}^s), \tag{15}$$

where $h(\cdot)$ denotes the measurement function associated with the supervisory signal. To obtain a reliable initialization for the propagated mean, we first optimize the least-squares objective

$$L_{\mathrm{MSE}}(\theta) = r^\top r. \tag{16}$$

This stage only aligns the propagated mean with the supervisory measurements.

Starting from this initialization, we then optimize the covariance-aware negative log-likelihood

$$L_{\mathrm{NLL}}(\theta) = r^\top \left(H\bar{\boldsymbol{P}}^s H^\top + W\right)^{-1} r + \log\det\left(H\bar{\boldsymbol{P}}^s H^\top + W\right), \tag{17}$$

where H is the Jacobian of the residual with respect to the supervisory-state error, and W is the supervisory measurement noise covariance. This weighting naturally accounts for accumulated propagation uncertainty, so measurements associated with longer or noisier propagation intervals are assigned smaller confidence.

5.3 Ground-Truth State Supervision

We first consider the case where ground-truth poses are available from an external motion-capture system. When needed, pseudo ground-truth velocities may be obtained by numerical differentiation, yielding a supervisory state sequence $\{X_k\}_{k=1}^N$.

Let $\bar{\boldsymbol{X}} = \{\bar{X}_1, \bar{X}_2, \ldots, \bar{X}_N\}$ denote the propagated trajectory. Let ξ_k be the invariant error associated with $\bar{X}_k$. After discretization of the invariant error dynamics, the lower-level uncertainty propagation admits the linear Gaussian model

$$\xi_{k+1} = F_k \xi_k + w_k, \qquad w_k \sim \mathcal{N}(0, Q_k), \tag{18}$$

with

$$\xi_1 \sim \mathcal{N}(0, P_1). \tag{19}$$

For each supervisory state, define the residual

$$r_k = X_k \boxminus \bar{X}_k. \tag{20}$$

Under the first-order approximation in invariant error coordinates,

$$r_k \approx H_k \xi_k, \tag{21}$$

where H_k is the corresponding linearization Jacobian.

5.3.1 Chain-Structured Likelihood

We first derive the supervisory objective in a chain-structured form. In the noiseless-supervision case, and assuming each H_k is invertible, define

$$z_k := H_k^{-1} r_k \approx \xi_k. \tag{22}$$

Then $\{z_k\}_{k=1}^N$ follows the Gaussian Markov model

$$z_{k+1} = F_k z_k + w_k, \qquad w_k \sim \mathcal{N}(0, Q_k), \tag{23}$$

with

$$z_1 \sim \mathcal{N}(0, P_1). \tag{24}$$

Therefore, the residual trajectory likelihood admits the factorization

$$p(r_{1:N} \mid \theta) = p(r_1 \mid \theta) \prod_{k=1}^{N-1} p(r_{k+1} \mid r_k, \theta). \tag{25}$$

More specifically,

$$p(r_1 \mid \theta) = \mathcal{N}\big(r_1;\, 0,\; H_1 P_1 H_1^\top\big), \tag{26}$$

and, for $k = 1, \dots, N-1$,

$$p(r_{k+1} \mid r_k, \theta) = \mathcal{N}\big(r_{k+1};\, H_{k+1} F_k H_k^{-1} r_k,\; H_{k+1} Q_k H_{k+1}^\top\big)\,. \tag{27}$$

Hence, up to an additive constant, the negative log-likelihood is

$$\begin{aligned} L_{\mathrm{gt}}(\theta) = \frac{1}{2} r_1^\top (H_1 P_1 H_1^\top)^{-1} r_1 + \frac{1}{2}\log\det(H_1 P_1 H_1^\top) \\ + \frac{1}{2}\sum_{k=1}^{N-1}\Big[\big(r_{k+1} - H_{k+1}F_k H_k^{-1} r_k\big)^\top (H_{k+1} Q_k H_{k+1}^\top)^{-1}\big(r_{k+1} - H_{k+1}F_k H_k^{-1} r_k\big) \\ + \log\det(H_{k+1} Q_k H_{k+1}^\top)\Big]. \end{aligned} \tag{28}$$

Equivalently, by defining

$$\epsilon_k := H_{k+1}^{-1} r_{k+1} - F_k H_k^{-1} r_k, \tag{29}$$

the same objective can be written as

$$L_{\mathrm{gt}}(\theta) = \frac{1}{2} r_1^\top (H_1 P_1 H_1^\top)^{-1} r_1 + \frac{1}{2}\log\det(H_1 P_1 H_1^\top) + \frac{1}{2}\sum_{k=1}^{N-1}\left[\epsilon_k^\top Q_k^{-1}\epsilon_k + \log\det(H_{k+1} Q_k H_{k+1}^\top)\right]. \tag{30}$$

This form shows that the supervisory objective is induced by the lower-level Markov dynamics, rather than by treating residuals at different time steps independently.

5.3.2 Equivalent Precision-Matrix Form

The same objective also admits an equivalent global Gaussian form, which is useful for exposing the sparsity structure inherited from lower-level propagation. Let

$$\boldsymbol{\xi} = [\xi_1^\top, \xi_2^\top, \dots, \xi_N^\top]^\top \tag{31}$$

denote the stacked invariant error vector. Since the dynamics is Markovian, $\boldsymbol{\xi}$ follows a joint Gaussian distribution with covariance P, whose precision matrix P^{-1} is block tridiagonal:

$$P^{-1} = \begin{bmatrix} \Lambda_{1,1} & \Lambda_{1,2} & 0 & \cdots & 0 \\ \Lambda_{2,1} & \Lambda_{2,2} & \Lambda_{2,3} & \ddots & \vdots \\ 0 & \Lambda_{3,2} & \Lambda_{3,3} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \Lambda_{N-1,N} \\ 0 & \cdots & 0 & \Lambda_{N,N-1} & \Lambda_{N,N} \end{bmatrix}, \tag{32}$$

where

$$\begin{aligned} \Lambda_{1,1} &= P_1^{-1} + F_1^\top Q_1^{-1} F_1, \\ \Lambda_{k,k} &= Q_{k-1}^{-1} + F_k^\top Q_k^{-1} F_k, \qquad k = 2, \dots, N-1, \\ \Lambda_{N,N} &= Q_{N-1}^{-1}, \\ \Lambda_{k,k+1} &= -F_k^\top Q_k^{-1}, \qquad k = 1, \dots, N-1 = \Lambda_{k+1,k}^\top. \end{aligned}$$

Now stack the residuals and Jacobians as

$$r = \begin{bmatrix} r_1^\top & r_2^\top & \cdots & r_N^\top \end{bmatrix}^\top, \qquad H = \mathrm{blkdiag}(H_1, \dots, H_N). \tag{33}$$

Neglecting supervisory noise for simplicity, one has

$$r \sim \mathcal{N}(0, \Sigma), \qquad \Sigma = H P H^\top. \tag{34}$$

Hence, up to an additive constant,

$$L_{\mathrm{gt}}(\theta) = r^\top \Sigma^{-1} r + \log\det\Sigma. \tag{35}$$

Since H is block diagonal and invertible,

$$\Sigma^{-1} = H^{-\top} P^{-1} H^{-1}, \tag{36}$$

and

$$\log\det\Sigma = \log\det P + 2\sum_{k=1}^{N} \log\left|\det H_k\right|. \tag{37}$$

Therefore,

$$L_{\mathrm{gt}}(\theta) = r^\top H^{-\top} P^{-1} H^{-1} r + \log\det P + 2\sum_{k=1}^{N} \log\left|\det H_k\right|. \tag{38}$$

This is equivalent to the chain-structured likelihood above. The chain form is convenient for probabilistic interpretation, whereas the precision-matrix form makes the sparsity structure explicit.

Computational remark. Although the covariance of the stacked residual trajectory is generally dense, the lower-level precision matrix P^{-1} remains block tridiagonal. Therefore, when the state dimension is fixed, the objective can be evaluated in linear time with respect to the trajectory length.

5.4 Loop-Closure Supervision

We next consider the case where the supervisory signal consists of relative-pose loop-closure measurements. Let the stacked loop-closure residual be modeled as

$$r = H\xi + v, \qquad v \sim \mathcal{N}(0, R), \tag{39}$$

where $\xi \sim \mathcal{N}(0, P)$ denotes the stacked lower-level trajectory error, R is the loop-closure noise covariance, and each row block of H only involves the two states associated with the corresponding loop closure. The marginal residual distribution is

$$r \sim \mathcal{N}(0, S), \qquad S = HPH^\top + R. \tag{40}$$

Although S is generally dense, its negative log-likelihood can still be evaluated by exploiting the sparse lower-level precision matrix P^{-1}. Define

$$K := P^{-1} + H^\top R^{-1} H, \qquad \eta := H^\top R^{-1} r. \tag{41}$$

By the Woodbury identity and the matrix determinant lemma,

$$S^{-1} = R^{-1} - R^{-1} H K^{-1} H^\top R^{-1}, \tag{42}$$

and

$$\log\det S = \log\det R + \log\det K - \log\det P^{-1}. \tag{43}$$

Therefore, up to an additive constant, the negative log-likelihood becomes

$$L_{\mathrm{lc}}(\theta) = \frac{1}{2} r^\top R^{-1} r - \frac{1}{2} \eta^\top K^{-1} \eta + \frac{1}{2}\log\det R + \frac{1}{2}\log\det K - \frac{1}{2}\log\det P^{-1}. \tag{44}$$

This shows that, although the marginal loop-closure covariance is dense, one can still exploit the chain-structured sparsity inherited from lower-level IMU propagation. Each loop closure only introduces an additional sparse long-range coupling between the two involved states.

6 Gradient Estimation

By the chain rule, the effect of θ flows from $b_k = \mathrm{vec}(b_a, b_g)$ through T_k to loss $\mathcal{L}$. The gradient computation follows several steps. First, consider the estimated trajectory error as an **implicit function of** b_k, then compute the gradient $\frac{\partial \mathcal{L}}{\partial b_k}$ through backpropagation as discussed in 6.1. Second, using the adjoint method to compute $\frac{\partial \mathcal{L}}{\partial \theta}$ with detail in 6.2.

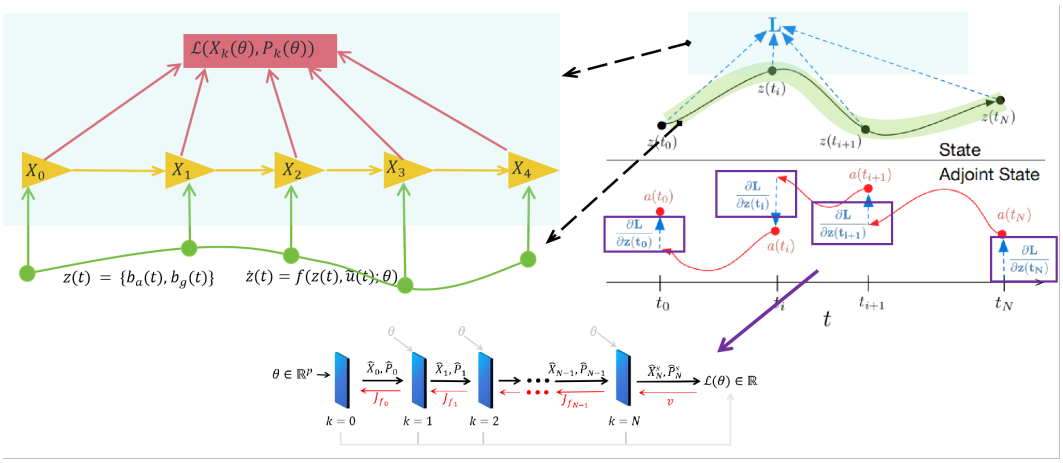


Figure 1: Modules of the algorithm.

6.1 Outer Adjoint for Bias Sensitivity

We analyze the sensitivity of the invariant error dynamics with respect to the IMU bias. Let $\xi_k \in \mathbb{R}^d$ denote the right-invariant error at time step k. Recall that $n_t = [n_t^a; n_t^g]$. Under first-order linearization, the error propagation admits the form

$$\xi_{k+1} = F_k \xi_k + G_k n_k,$$

where F_k and G_k are the state transition and noise Jacobians. By adopting the right-invariant error, the continuous-time dynamics in (6) become state-trajectory independent, and A_t is constant:

$$A_t = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ g^\wedge & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I & \mathbf{0} \end{bmatrix}.$$

According to the linearization of the IMU kinematics, G_k and F_k take the form

$$G_k = -\Delta t_k \begin{bmatrix} 0 & \bar{R}_k \\ \bar{R}_k & (\bar{v}_k)^\wedge \bar{R}_k \\ 0 & (\bar{p}_k)^\wedge \bar{R}_k \end{bmatrix}, \qquad F_k = \exp(A_t \Delta t_k) = \begin{bmatrix} I & 0 & 0 \\ g^\wedge \Delta t_k & I & 0 \\ \frac{1}{2} g^\wedge \Delta t_k^2 & I \Delta t_k & I \end{bmatrix}.$$

To optimize the bias, the error system of interest is

$$\xi_{k+1} = F_k \xi_k + M_k \delta b_k. \tag{45}$$

Since the noise term depends explicitly on the bias

$$b_k = \begin{bmatrix} b_k^a \\ b_k^g \end{bmatrix},$$

and we maintain the ordering

$$n_k = \begin{bmatrix} \tilde{a}_k - R_k^\top (a_k - g) - b_k^a \\ \tilde{\omega}_k - \omega_k - b_k^g \end{bmatrix}, \qquad \frac{\partial n_k}{\partial b_k} = -I,$$

it follows that

$$M_k = G_k \frac{\partial n_k}{\partial b_k} = -G_k.$$

Note that ξ_k depends only on the past biases $b_{0:k-1}$ and is independent of b_k.

We consider a stage-wise loss defined on the invariant error:

$$L = \sum_{k=1}^{N} \ell_k(\xi_k). \tag{46}$$

Define the adjoint variable as

$$\lambda_k \triangleq \frac{\partial L}{\partial \xi_k}, \qquad \lambda_N = \frac{\partial \ell_N}{\partial \xi_N}. \tag{47}$$

Applying the chain rule to (45), the adjoint satisfies the backward recursion

$$\lambda_k = \frac{\partial \ell_k}{\partial \xi_k} + F_k^\top \lambda_{k+1}. \tag{48}$$

Finally, the gradient of the total loss with respect to the bias perturbation is

$$\frac{\partial L}{\partial \delta b_k} = \frac{\partial L}{\partial \xi_{k+1}} \frac{\partial \xi_{k+1}}{\partial \delta b_k} = M_k^\top \lambda_{k+1} = -G_k^\top \lambda_{k+1}. \tag{49}$$

Remark. While neural ODE frameworks are typically developed in Euclidean spaces and rely on automatic differentiation through nonlinear dynamics, directly applying them to Lie group systems is nontrivial due to the manifold structure. Instead of differentiating through the nonlinear IMU kinematics on $\mathrm{SE}_2(3)$, we leverage the group-affine property of the system. By introducing the invariant error $\xi = \log(\eta)$, the error dynamics admit a log-linear representation. This allows us to construct a linear surrogate system in the Lie algebra, through which gradients can be propagated efficiently and in closed form.

6.2 Inner Adjoint Method for NODE Training

We have modeled the bias dynamics as NODE with network parameter θ. To optimize θ, the b_k can be considered as the latent variables. After computing the $\frac{\partial L}{\partial b_k}$ in the subsection 6.1, we can use adjoint method to compute the gradient $\frac{\partial L}{\partial \theta}$ as described in 3.4.

Also, there are other variants for improving the stability of the NODE training as introduced in the following context.

7 Experimental Results

The integration path should be relatively short. The motion pattern should have diversity.

7.1 Trajectory Patterns

8 Conclusion

A Bias to Invariant Error to Loss

References

[1] A. Barrau and S. Bonnabel. The invariant extended kalman filter as a stable observer. *IEEE Transactions on Automatic Control*, 62(4):1797–1812, 2016.

[2] B. Liu, T.-Y. Lin, W. Zhang, and M. Ghaffari. Debiasing 6-dof imu via hierarchical learning of continuous bias dynamics. In *Robotics: Science and Systems (RSS)*, 2025.